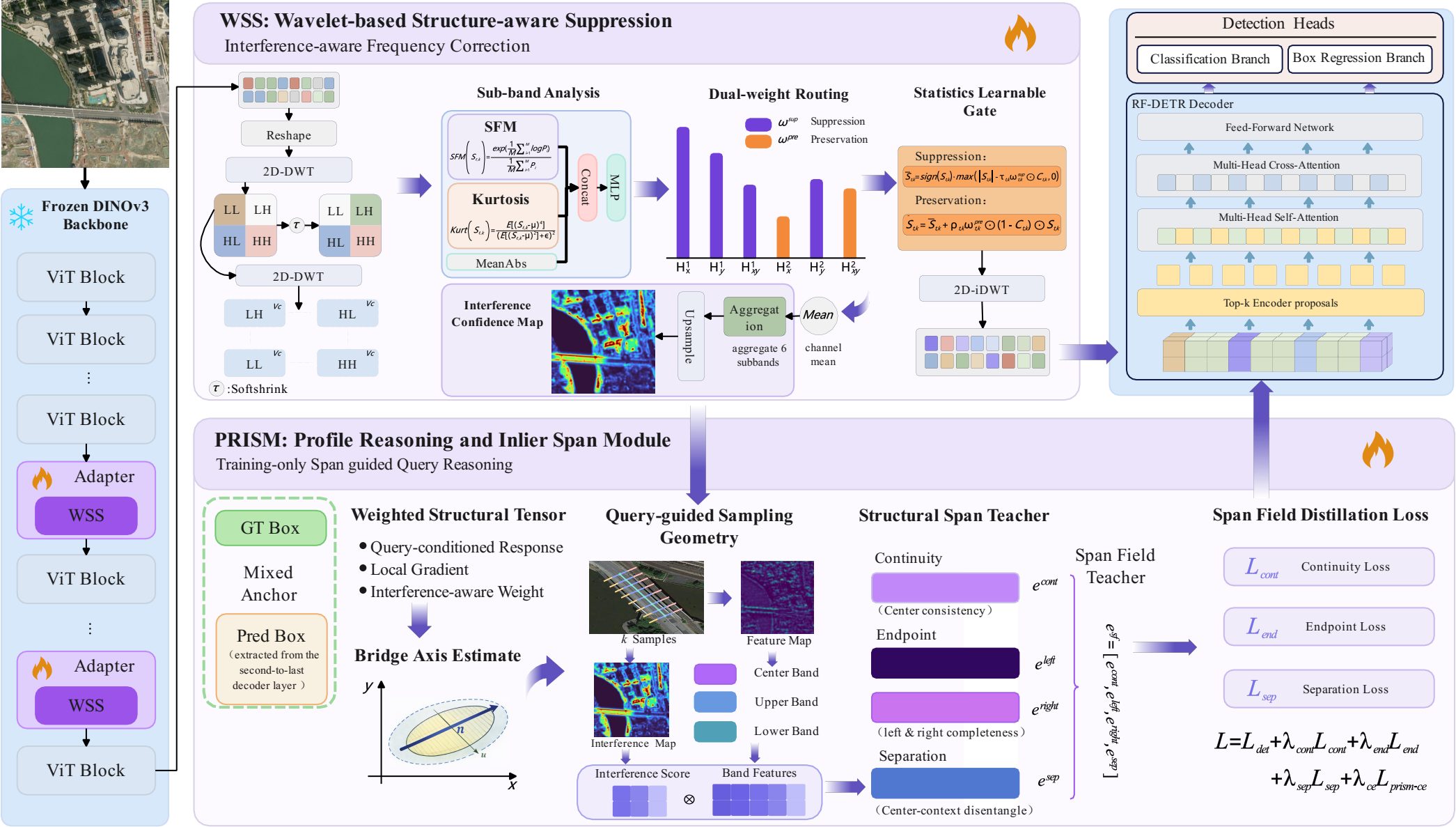
WSS: Wavelet-based Structure-aware Suppression
Interference-aware Frequency Correction
Frozen DINOv3 Backbone
ViT Block
ViT Block
ViT Block
Adapter
WSS
ViT Block
Adapter
WSS
ViT Block
Reshape
2D-DWT
LL LH HL HH
2D-DWT
LH HL LL HH
:Softshrink
Sub-band Analysis
SFM
Kurtosis
MeanAbs
Concat
MLP
Dual-weight Routing
Suppression
Preservation
Statistics Learnable Gate
Suppression:
Preservation:
2D-iDWT
Interference Confidence Map
Upsample
Aggregation
aggregate 6 subbands
Mean
channel mean
Detection Heads
Classification Branch
Box Regression Branch
RF-DETR Decoder
Feed-Forward Network
Multi-Head Cross-Attention
Multi-Head Self-Attention
Top-k Encoder proposals
PRISM: Profile Reasoning and Inlier Span Module
Training-only Span guided Query Reasoning
GT Box
Mixed Anchor
Pred Box
(extracted from the second-to-last decoder layer)
Weighted Structural Tensor
Query-conditioned Response
Local Gradient
Interference-aware Weight
Bridge Axis Estimate
Query-guided Sampling Geometry
k Samples
Feature Map
Interference Map
Center Band
Upper Band
Lower Band
Interference Score
Band Features
Structural Span Teacher
Continuity
(Center consistency)
Endpoint
(left & right completeness)
Separation
(Center-context disentangle)
Span Field Teacher
Span Field Distillation Loss
Continuity Loss
Endpoint Loss
Separation Loss